

Research Notes on DALL-E image generation

DALL-E image generation

>> Section 1

History and background DALL-E was revealed by OpenAI in a blog post on 5 January 2021, and uses a version of GPT-3 modified to generate images. On 6 April 2022, OpenAI announced DALL-E 2, a successor designed to generate more realistic images at higher resolutions that "can combine concepts, attributes, and styles". On 20 July 2022, DALL-E 2 entered into a beta phase with invitations sent to 1 million waitlisted individuals; users could generate a certain number of images for free every month and may purchase more. Access had previously been restricted to pre-selected users for a research preview due to concerns about ethics and safety. On 28 September 2022, DALL-E 2 was opened to everyone and the waitlist requirement was removed. In September 2023, OpenAI announced their latest image model, DALL-E 3, capable of understanding "significantly more nuance and detail" than previous iterations. In early November 2022, OpenAI released DALL-E 2 as an API, allowing developers to integrate the model into their own applications. Microsoft unveiled their implementation of DALL-E 2 in their Designer app and Image Creator tool included in Bing and Microsoft Edge. The API operates on a cost-per-image basis, with prices varying depending on image resolution. Volume discounts are available to companies working with OpenAI's enterprise team. The software's name is a portmanteau of the names of animated robot Pixar character WALL-E and the Spanish surrealist artist Salvador Dalí. In February 2024, OpenAI began adding watermarks to DALL-E generated images, containing metadata in the C2PA (Coalition for Content Provenance and Authenticity) standard promoted by the Content Authenticity Initiative.

>> Section 2

Ramesh, Aditya; Pavlov, Mikhail; Goh, Gabriel; Gray, Scott; Voss, Chelsea; Radford, Alec; Chen, Mark; Sutskever, Ilya (26 February 2021). "Zero-Shot Text-to-Image Generation". arXiv:2102.12092 [cs.CV].. The original report on DALL-E. DALL-E 3 System Card DALL-E 3 paper by OpenAI DALL-E 2 website Craiyon website

>> Section 3

Most coverage of DALL-E focuses on a small subset of "surreal" or "quirky" outputs. DALL-E's output for "an illustration of a baby daikon radish in a tutu walking a dog" was mentioned in pieces from Input, NBC, Nature, and other publications. Its output for "an armchair in the shape of an avocado" was also widely covered. ExtremeTech stated "you can ask DALL-E for a picture of a phone or vacuum cleaner from a specified period of time, and it understands how those objects have changed". Engadget

also noted its unusual capacity for "understanding how telephones and other objects change over time". According to MIT Technology Review, one of OpenAI's objectives was to "give language models a better grasp of the everyday concepts that humans use to make sense of things". Wall Street investors have had a positive reception of DALL-E 2, with some firms thinking it could represent a turning point for a future multi-trillion dollar industry. By mid-2019, OpenAI had already received over \$1 billion in funding from Microsoft and Khosla Ventures, and in January 2023, following the launch of DALL-E 2 and ChatGPT, received an additional \$10 billion in funding from Microsoft. Japan's anime community has had a negative reaction to DALL-E 2 and similar models. Two arguments are typically presented by artists against the software. The first is that AI art is not art because it is not created by a human with intent. "The juxtaposition of AI-generated images with their own work is degrading and undermines the time and skill that goes into their art. AI-driven image generation tools have been heavily criticized by artists because they are trained on human-made art scraped from the web." The second is the trouble with copyright law and data text-to-image models are trained on. OpenAI has not released information about what dataset(s) were used to train DALL-E 2, inciting concern from some that the work of artists has been used for training without permission. Copyright laws surrounding these topics are inconclusive at the moment. After integrating DALL-E 3 into Bing Chat and ChatGPT, Microsoft and OpenAI faced criticism for excessive content filtering, with critics saying DALL-E had been "lobotomized." The flagging of images generated by prompts such as "man breaks server rack with sledgehammer" was cited as evidence. Over the first days of its launch, filtering was reportedly increased to the point where images generated by some of Bing's own suggested prompts were being blocked. TechRadar argued that leaning too heavily on the side of caution could limit DALL-E's value as a creative tool.

>> Section 4

Technology The first generative pre-trained transformer (GPT) model was initially developed by OpenAI in 2018, using a Transformer architecture. The first iteration, GPT-1, was scaled up to produce GPT-2 in 2019; in 2020, it was scaled up again to produce GPT-3, with 175 billion parameters.

>> Section 5

Ethical concerns DALL-E 2's reliance on public datasets influences its results and leads to algorithmic bias in some cases, such as generating higher numbers of men than women for requests that do not mention gender. DALL-E 2's training data was filtered to remove violent and sexual

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imagery, but this was found to increase bias in some cases such as reducing the frequency of women being generated. OpenAI hypothesise that this may be because women were more likely to be sexualised in training data which caused the filter to influence results. In September 2022, OpenAI confirmed to The Verge that DALL-E invisibly inserts phrases into user prompts to address bias in results; for instance, "black man" and "Asian woman" are inserted into prompts that do not specify gender or race. OpenAI claims to address concerns for potential "racy content" – containing nudity or sexual content generation, with DALL-E 3 through input/output filters, blocklists, ChatGPT refusals, and model level interventions. However, DALL-E 3 continues to disproportionally represent people as White, female, and youthful. Users are able to somewhat remedy this through more specific prompts for image generation. A concern about DALL-E 2 and similar image generation models is that they could be used to propagate deepfakes and other forms of misinformation. As an attempt to mitigate this, the software rejects prompts involving public figures and uploads containing human faces. Prompts containing potentially objectionable content are blocked, and uploaded images are analysed to detect offensive material. A disadvantage of prompt-based filtering is that it is easy to bypass using alternative phrases that result in a similar output. For example, the word "blood" is filtered, but "ketchup" and "red liquid" are not. Another concern about DALL-E 2 and similar models is that they could cause technological unemployment for artists, photographers, and graphic designers due to their accuracy and popularity. DALL-E 3 is designed to block users from generating art in the style of currently-living artists. While OpenAI states that images produced using these models do not require permission to reprint, sell, or merchandise, legal concerns have been raised regarding who owns those images. In 2023 Microsoft pitched the United States Department of Defense to use DALL-E models to train battlefield management systems. In January 2024 OpenAI removed its blanket ban on military and warfare use from its usage policies.